

The role of parasitoids in a nursery-pollinator system: A population dynamics model

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ABSTRACT

A nursery pollinator system comprises a plant and an insect pollinator which uses the reproductive structures of the plant to guard its eggs and serve it as food for the larvae. This kind of systems usually includes a third party, one or more species of parasitoids, that inject their eggs inside the larvae of the pollinating-seed predator to feed on them from the inside out. Here we build a model based on differential equations that replicates this system, first by showing the dual role of the pollinating-seed predator, which behaves both as a mutualist and as an antagonist for the plant. We show that the system is more stable with the presence of the parasitoids; the stationary solutions with them significantly favor the plant population size with minor effects for the population of the pollinating-seed predator. By modeling both sexes of the nursery pollinator separately, we also highlight the role that male adults can have to compensate the costs and benefits of the interaction with plants, and thus to contribute to the stability of such mutualisms.

1. Introduction

Probably the most paradigmatic mutualistic system is pollination (Toby Kiers et al., 2010; Holland et al., 2002), in which the flower visitors obtain trophic resources (nectar, pollen, oils, etc.) and the plant obtains reproductive benefits through pollen dispersal and seed production. A special case of pollination, called nursery pollination, is reported when one pollinator species uses the fertilized flowers as hatching nests for its eggs, and the developing seeds serve as a food resource for its larvae. Therefore, the pollinating-seed predator obtains plant food resources for both its adult and larval stages, depriving the plant of part of the reproductive benefits obtained in a legitimate pollination interaction (Bronstein, 2001). Although this kind of interaction sounds eccentric, nursery pollination had been described in several families of angiosperms, with classic examples like fig trees-fig wasps and yuccas-yucca moths (Dufay and Anstett, 2003). In these systems, there must exist a subtle balance between the benefits (pollination) and costs (seed predation) to achieve a positive net outcome to the plant. Indeed, the balance may shift from positive to negative, and therefore from mutualism to antagonism, depending on the mechanisms evolved by the hosts to avoid over-exploitation and on the ecological context where the interaction takes place (Leung and Poulin, 2008).

One mechanism controlling over-exploitation by the plant consists

of random abortion of fruits, or selective abortion of fruits containing eggs or larvae of the pollinating-seed predator. This feature has been argued, by mathematical modeling approaches, as an evolutionary strategy of interspecific population regulation (Holland and DeAngelis, 2001, 2006; Westerbergh and Westerbergh, 2001). With respect to the ecological context, apart from regulation of population densities due to environmental fluctuations, a central topic is related to the presence of third partners. One central consideration is whether the seed predator is also the unique pollinator or not (hereafter, obligate and facultative nursery pollination systems, respectively). In facultative systems, the occurrence of co-pollinators can shift the sign of this interaction, because they contribute to seed production without the cost of seed predation (Thompson and Pellmyr, 1992). The nursery pollination system may be also affected by the existence of predators, pathogens and parasitoids that may regulate the population size of the pollinating-seed predator, avoiding excess seed loss and system instability. Several field studies suggest this kind of regulatory process in nursery pollination systems (Force and Thompson, 1984; Elzinga et al., 2003; Crabb and Pellmyr, 2006; Nunes et al., 2018), but it has not been tested by population dynamic models.

Complex biological systems such as the nursery pollination are paradigmatic frameworks because different actors can play a vital role in the equilibrium of the ecological system, and the importance of an

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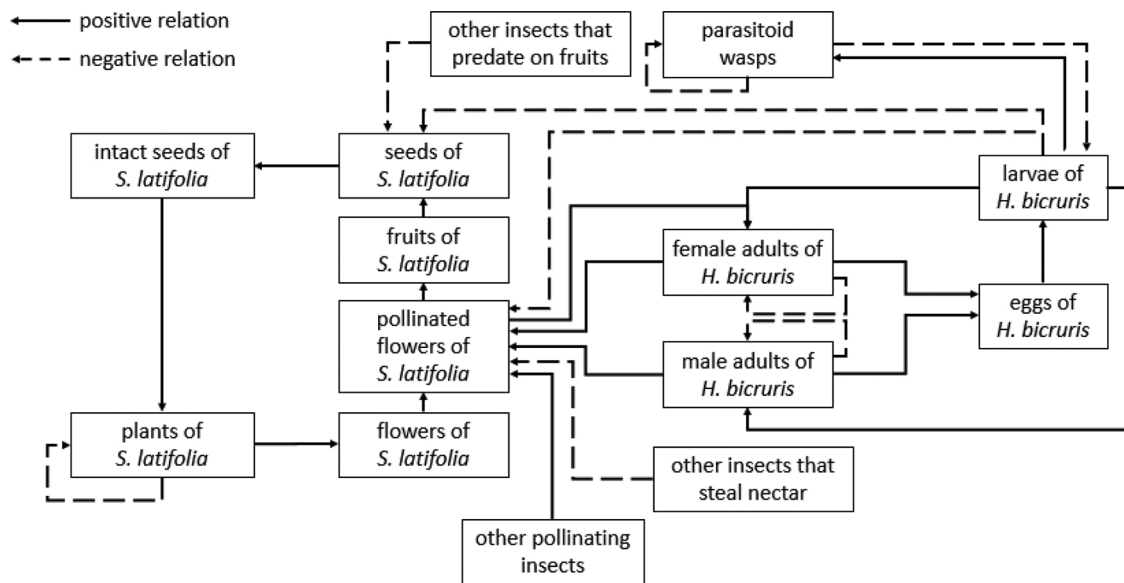


Fig. 1. Conceptual diagram of the system. Mutualistic interactions are represented by solid lines and antagonistic relations by dashed lines. Intraspecific interactions are considered as competition, so they are also represented by dashed lines. Only direct relations are considered.

apparently secondary species can be addressed using population dynamic modeling (Neuhauser and Fargione, 2004; Holland et al., 2002). In this study, we use a generalized version of the model of García-Algarra et al. (2014), which can describe multiple ecological interactions within the same equations. This approach has the advantage of treating species interactions based on fundamental ecological laws (Turchin, 2003) and independently of the model, i.e. without having to consider them as different phenomena, which is a conceptual burden. Such a generalization may allow researchers a better representation of ecological systems by including most observed interactions, weighting their importance where necessary and avoiding frameworks of specific systems and ad hoc modeling.

We hypothesize that, when the population density of two species can alternate the net effect of their interaction between mutualism and parasitism, the presence of a third species can regulate their population dynamics and stabilize the interaction towards mutualism. To test this prediction, we applied a generalized model of ecological relations to the system made up of Caryophyllaceous plants, the nursery pollinator *Hadena* moths and their parasitoid wasps. We expect that a subtle regulation of the moths population by its parasitoids may stabilize the dynamics of the system, allowing the high metabolic costs of other regulatory processes, like fruit abortion, to control over-exploitation.

2. Materials and methods

We first describe the ecological system studied to build a conceptual diagram that we used to model the system with a set of differential equations.

2.1. The ecological system

Some members of the plant family Caryophyllaceae, especially the genus *Silene*, are involved in a facultative nursery pollination interaction with moths of the genus *Hadena* (Noctuidae). These plants have many diurnal and nocturnal flower visitors, but *Hadena* moths are among their most effective pollinators (Kephart et al., 2005; Prieto-Benítez et al., 2017). Male moths only pollinate the flowers, but female moths pollinate and also lay their eggs on fertilized flowers (Labouche and Bernasconi, 2010). Eggs successfully hatched give rise to the larvae, which penetrate into the primary fruits and feed on the seeds. Once the primary fruit is consumed, the larvae feed on a variable number of

secondary fruits to complete their development. A recent review (Prieto-Benítez et al., 2017) confirmed nursery pollination between 22 *Hadena* species and 70 Caryophyllaceae species, but most of the information on this system comes from the best-studied pair of interacting species, *S. latifolia* and *H. bicruris*. Extensive surveys found that *H. bicruris* occurred in more than 90% of European *S. latifolia* populations and consumed 25–70% of annual fruit production (Wolfe, 2002).

Other species are involved in this system (Prieto-Benítez et al., 2017). Some of the multiple diurnal and nocturnal flower visitors are legitimate pollinators that have only a positive effect on the plant, but other species (some bees and bumblebees) may behave as nectar robbers, which may affect the fertility of the plant by reducing this flower resource for the legitimate pollinators. Second, a few other insects predate flowers and fruits and therefore constitute a menace to the plants and competitors to the moths. Third, *Hadena* moths have some parasitoids, mainly from the Braconidae and Ichneumonidae wasp families (Elzinga et al., 2003). After feeding on the primary fruit, *Hadena* caterpillars are too large to be protected within the fruits they consume, so they are prone to attack by these parasitoids, which inject their eggs inside them. When these eggs hatch, the larvae feed on the host fluids without causing immediate death. *Hadena* larvae continue feeding on the plant's fruits until growth is complete, and then the larvae of parasitoids emerge from the host's body, killing it. At least 13 parasitoid species are associated with larvae of *H. bicruris*, but the best studied species is the braconid *Microplitis tristis* (Elzinga et al., 2007). These parasitoids may act as a control mechanism for the *Silene*-*Hadena* system, as demonstrated later. Finally, another organism may be involved in the system, the obligate fungus pathogen *Microbotryum*. It is a complex of cryptic species infecting many Caryophyllaceous species, which replaces the pollen with fungal spores that are spread by pollinating insects (Antonovics et al., 2002). *Microbotryum* infestation causes the sterilization of anthers and also reduce the size of flowers (Shykoff and Kaltz, 1998). *Hadena* moths show an adaptive response, avoiding diseased plants to oviposit (Biere and Honders, 1996).

Fig. 1 shows the interactions among all these species, although we simplified the system for modeling purpose (Fig. 2). We do not consider *Microbotryum*, since its presence regulates the interaction between plants and moths and that would require a third order term in the interactions. Moreover, fungal population strictly depends on plant population, so it cannot be included neither as a net effect on plant independent parameters, as we do with other pollinating and predatory

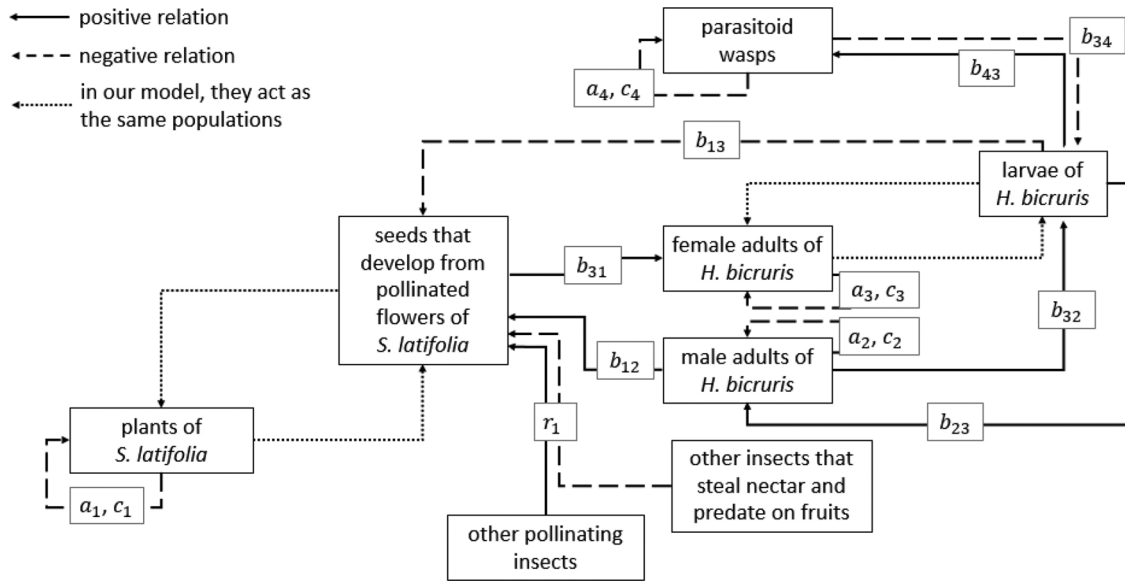


Fig. 2. Conceptual diagram of the reduced system. Mutualistic interactions are represented by solid lines, and antagonistic interactions are represented by dashed lines. Dotted lines indicate species that have multiple developmental stages but are treated as a single stage. Labels on the interaction lines indicate coefficients in the set of differential equations.

insects. Therefore, the effect of *Microbotryum* on the population dynamics of the system require a more complex approach that could be explored in subsequent studies.

2.2. The reduced system and translation into equations

Our ecological system consist of many interactions that were simplified to make a coherent model that we later translated into a set of differential equations. We followed this approach:

- 1 We consider only the interactions between one Caryophyllaceae (X_1), males and females of one *Hadena* species (X_2 and X_3 , respectively), and a third species, moth parasitoid (X_4), which predate *Hadena*.
- 2 The interaction between *Hadena* and Caryophyllaceae are divided between male and female moths. Male moths act as obligate mutualists for both the Caryophyllaceae and the female moths. Female moths act as facultative mutualists for the male moths but as predators for the Caryophyllaceae, since moth adults and larvae stages are pooled in a single stage. The population of larvae was estimated by the difference $larvae = X_3 - X_2$, or $adults = X_2 + X_3$ if $X_2 > X_3$.
- 3 All moth larvae are assumed to be female. Therefore, male moths are generated by a mutualistic term that represents the interaction between males and females, modulated by coefficient b_{23} .
- 4 Caryophyllaceae is assumed to be sex neutral. Although this can be justified by the existence of many monoecious subgenera, dioecious species can also be modeled assuming only one sex. We demonstrate this in A, but the main reason is that male and female plants in this system differ only in the intraspecific interactions, which do not affect the relations with moths and moth parasitoids.
- 5 Parasitoids are modeled as predators and interact only with female moths, since females also include the larval stage. The latter interaction is modulated by two coefficients: b_{34} and b_{43} . Although we modeled only one parasitoid species, the impact of parasitoids may not be reduced to only one species, so we considered that b_{34} , the influence of parasitoids on female moths and their larvae, can be weighted by a factor that considers additional species. We discuss this in detail in B.

To represent the reduced system (Fig. 2), we used a generalized version of the logistic-mutualistic system of García-Algarra et al. (2014), which can be summarized, for n species, as:

$$\frac{\dot{X}_i}{X_i} = r_i + \sum_{k=1}^n b_{ik} X_k - \left(a_i + c_i \sum_{k=1}^n b_{ik} X_k \right) X_i. \quad (1)$$

The set of equations that reflect our assumptions in this system is:

$$\frac{\dot{X}_1}{X_1} = (r_1 + b_{12} X_2 + b_{13} X_3) - (a_1 + c_1 (b_{12} X_2 + b_{13} X_3)) X_1, \quad (2)$$

$$\frac{\dot{X}_2}{X_2} = (r_2 + b_{23} X_3) - (a_2 + c_2 b_{23} X_3) X_2, \quad (3)$$

$$\frac{\dot{X}_3}{X_3} = (r_3 + b_{31} X_1 + b_{32} X_2 + b_{34} X_4) - (a_3 + c_3 (b_{31} X_1 + b_{32} X_2 + b_{34} X_4)) X_3, \quad (4)$$

$$\frac{\dot{X}_4}{X_4} = (r_4 + b_{43} X_3) - (a_4 + c_4 b_{43} X_3) X_4. \quad (5)$$

In this system, parameters r_i represent the intrinsic growth rate of each species i , parameters b_{ij} represent the interaction between species i and j , and parameters a_i and c_i represent two kinds of negative effects for species i . a_i is the saturation rate caused by intraspecific competition, and c_i represents the limitations caused by the occupation rate by members of the same species. The latter has a similar effect as intraspecific competition, not with respect to the environment but to interactions with the other species.

To clarify the meaning of all the parameters, we present them one by one. To estimate the value of each parameter, we performed a comprehensive review of the ecological literature of the system, as presented in B.

- r_i : represents the intrinsic growth rate of each species i . We assume that all $r_i < 0$, since species depends exclusively on their interactions to reproduce. Male moths need female moths, which in turn need male moths as well as the plants, since the female moths also include the larvae. Parasitoids can survive only in the presence of female moths, since they inject eggs inside moths' larvae. Of all the r_i , only r_1 can be considered positive as well as negative. But the only difference when $r_1 > 0$, besides an increase in the population size

(shown later), is that the plants can survive without any of the other species, which does not reflect the system we care to study. r_i is measured in units of time^{-1} , and its inverse is the mean lifetime of species i .

- b_{12} : represents the interaction between the plants and the male moths. Since it summarizes the entire interaction, it thus reflects how many viable seeds are born, for each fruit that matures, for each flower that a male moth pollinates, averaged by the mean number of flowers that a plant produces, per unit of time. At the end, it equals the number of new plants a male moth generates, on average, per unit of time. Since it measures plant births, $b_{12} > 0$ and is measured in units of $(\text{time.moths})^{-1}$.
- b_{13} : represents the interaction between the plants and both the female adult moths and larvae. We set $b_{13}^+ - b_{13}^-$, in which b_{13}^+ corresponds to the pollination effect of the female moths, a mutualistic interaction, and b_{13}^- represents the predatory behavior of the moth's larvae, an antagonistic interaction. In this case, we consider that $b_{13}^+ < b_{13}^-$, so the overall interaction is antagonistic for the plant. Thus, it reflects how many viable seeds die (instead of being born) for each fruit that matures, for each flower that a female moth pollinates, averaged by the mean number of flowers that a plant produces, per unit of time. Since it represents deaths, $b_{13} < 0$ and is measured in units of $(\text{time.moths})^{-1}$.
- b_{23} : since male moths reproduce only through female moths, this parameter represents how many of the viable larvae produced by a pairing between male and female moths are males, per unit of time. To calculate this number, it is necessary to know the average sex ratio. Therefore, b_{23} has units of $(\text{time.moths})^{-1}$ and, necessarily, $b_{23} > 0$.
- b_{31} : has a deeper meaning. Since we assumed that moths need plants to lay their eggs, and only female moths can do that, this parameter reflects how many viable larvae may arise, on average, from each plant, considering that the larvae eat mainly seeds, but also remains of the fruits and even flowers. A plant whose flowers and fruits are eaten cannot survive or reproduce and therefore dies. To measure this value, we calculated how many fruits and flowers an individual larva needs to complete its development and become an adult moth. This number reflects how many plants need to die to produce moths, per unit of time. b_{31} has units of $(\text{time.plants})^{-1}$ and is > 0 .
- b_{32} : means the same as b_{23} , but with respect to the female moth population. Therefore, it follows that $b_{32} > 0$.
- b_{34} : is the number of female moths that die for each individual parasitoid, per unit of time. We counted the female moths only since we treated the female moths and their larvae as a single stage. Therefore, b_{34} has units of $(\text{time.parasitoids})^{-1}$ and is < 0 .
- b_{43} : is the number of parasitoids born for each female moth that dies, per unit of time. It equals the inverse of the number of moths that a parasitoid needs to grow from egg to larva and then, to an adult. b_{43} has units of $(\text{time.moths})^{-1}$ and is > 0 .
- a_i : represents the limitation imposed by intraspecific competition (i.e. how much the survival of an individuals is reduced because of the presence of other individuals of the same species). In the absence of interactions, the inverse of this number represents the carrying capacity of a species over the area under study, times its mean lifetime. a_i is measured in units of $(\text{time.population})^{-1}$ and is always > 0 .
- c_i : even though this parameter is similar to a_i , it has a different meaning. c_i represents the relative presence of individuals of a species in the area under study, so it measures the competition from trophic interactions. We calculated its value by dividing the mean area that an individual occupies by the total area that it can grow into. Thus, c_i is a dimensionless parameter and always > 0 .
- $b_{14}, b_{21}, b_{24}, b_{41}, b_{42}$: are all 0. This happens because there are no significant direct interactions between plants and parasitoids, male moths and plants (even male moths feed on plants, they reproduce only through female moths), and male moths and parasitoids (the

latter only inject eggs inside the larvae, here represented as females only).

2.3. Solving the set of equations

We solved the system numerically using a set of ordinary differential equations (ODEs). We used Mathematica for the ODEs and figures, and Python for the global sensitivity analysis. ODEs were solved numerically switching between explicit and implicit Runge-Kutta methods when stiffness was detected. Since the system could not be solved analytically, we calculated stationary solutions, which represent stationary populations when the values are zero or any positive real number. We used the Jacobian determinant of the system, evaluated at the stationary solutions, to calculate their eigenvalues and to explore if the corresponding populations were either stable, unstable or saddle points. The initial conditions and numerical values of parameters used for the simulations are shown in B.

2.4. Global sensitivity analysis

We performed a global sensitivity analysis of our system to study how the uncertainty in the model output can be apportioned to different sources of uncertainty in the inputs. We have chosen a Sobol sensitivity analysis, using the SALib python library (Herman and Usher, 2017). Sobol's method is designed to determine how much of the variability in the model output depends upon each of the input parameters, either upon a single parameter or upon an interaction between different parameters (Sobol, 2001; Saltelli, 2002; Saltelli et al., 2010).

3. Results

3.1. The three populations system (no parasitoids)

By setting $b_{34} = 0$ and $b_{43} = 0$, we evaluated how the system behaves in the absence of parasitoids. In this case, the plant-nursery pollinator system had only two stable solutions, which might oscillate at first but stabilized later. One stable solution was obvious -extinction of all the populations-, and the other was a stable coexistence of plants, male moths and female moths (Fig. 3), in which the plant population declined to almost the extinction, before arose again. Even though there were solutions that corresponded to extinctions of some of the populations, all of them were unstable for the system, since they have an obligate dependence between them.

With $r_1 > 0$, there is a stable solution with a positive population of plants and either the extinction of male or both populations of moths. We do not present either case because they are of no interest for our

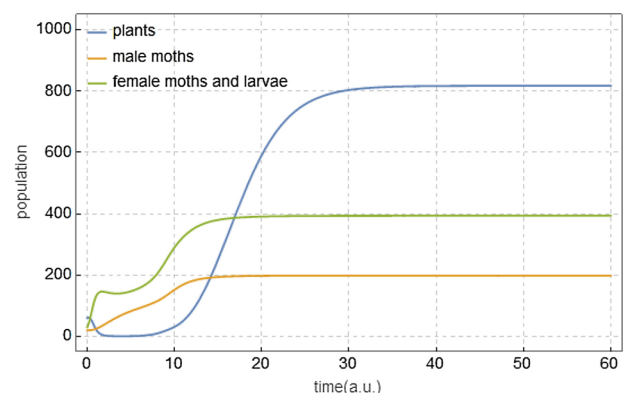


Fig. 3. Population dynamics with only three populations (plants and both male and female moths) and no parasitoids. Initial populations were set to $X_1(0) = 60$, $X_2(0) = 20$, $X_3(0) = 30$. Stationary stable populations are $X_1 = 817$, $X_2 = 198$, $X_3 = 393$.

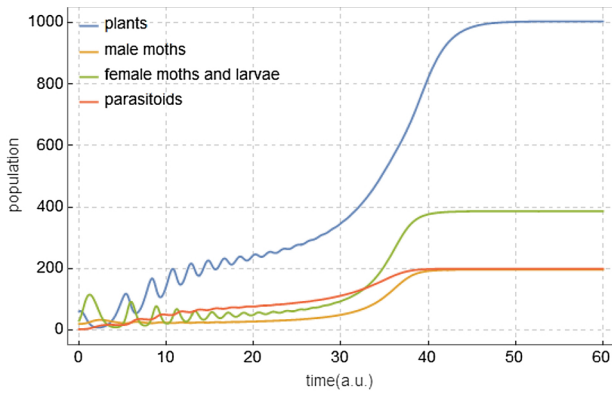


Fig. 4. Population dynamics with all the populations: plants, both male and female moths and parasitoids. Initial populations were set to $X_1(0) = 60$, $X_2(0) = 20$, $X_3(0) = 30$ and $X_4(0) = 2$. Stationary stable populations are $X_1 = 1004$, $X_2 = 196$, $X_3 = 387$, $X_4 = 199$.

system, as it depicts a scenario without moths.

3.2. The four populations system: the role of parasitoids

Although in the previous case the stationary solution of the system was stable when it included only three populations, this solution turned unstable when we introduced a fourth population, the parasitoid species; the solution became a saddle point. The set of four eigenvalues had three negative values and one positive value: $\lambda_i = \{-33.16, 4.26, -2.15, -0.31\}$. The X_4 axis of the four-dimensional phase space had an unstable stationary point on the extinction of the parasitoids. Therefore, the system presented only one stable solution, besides total extinction, corresponding to the coexistence of the four populations, with eigenvalues $\lambda_i = \{-15.56, -4.27, -2.54, -0.50\}$. The system oscillated at the beginning but stabilized later, and favored plant population compared to the system without parasitoids (Fig. 4). Even the presence of a very small population of parasitoids was enough to shift the system to a new stable stationary point, in which plant population increased notably (from $X_1 = 817$ to $X_1 = 1004$) with a negligible decrease in male and female moth populations. There was also an intermediate phase between $t = 20$ and $t = 30$, in which populations moved to a stationary unstable point, corresponding to $X_1 = 191$, $X_2 = 20$, $X_3 = 38$, $X_4 = 62$. The eigenvalues corresponding to this unstable solution had imaginary parts ($\lambda_i = \{-0.17 + 2.92i, -0.17 - 2.92i, -0.31, 0.081\}$), the reason why populations oscillate through them.

As mentioned, if $r_1 > 0$, i.e. by considering other non-nursery pollinators, the model predicted a notable increase in plant population (from $X_1 = 1090$ to $X_1 = 1200$, when $r_1 = 0.0973$), with a negligible effect in moth and parasitoid populations (Fig. 5).

The predation of the parasitoids on the moths, mediated by b_{34} ,

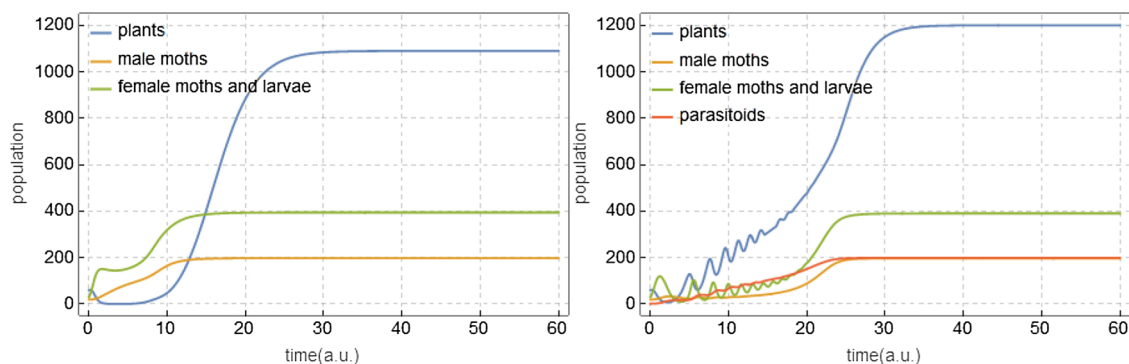


Fig. 5. Population dynamics of the system with additional pollinators, without parasitoids (left) and with parasitoids (right, $X_4(0) = 2$). Initial populations were set to $X_1(0) = 60$, $X_2(0) = 20$, $X_3(0) = 30$. Stationary stable populations are $X_1 = 1200$, $X_2 = 198$, $X_3 = 391$ and $X_4 = 200$.

regulates the effect of initial populations of parasitoids on the system (Fig. 6). When $b_{34} = -0.03$, which is 0.25 times the value we used before ($b_{34} = -0.122$), moths and parasitoid stationary stable populations remained almost the same (they changed less than 1.5 %), while for plants varied significantly (around 15 %). On the other hand, when $b_{34} = -0.183$, which is 1.5 times the value we used first, all populations became extinct, even though a stationary stable solution with $X_1 = 1166$, $X_2 = 195$, $X_3 = 380$ and $X_4 = 199$ remained. For the system to go to the latter solution, bigger initial populations were needed.

In this work, we have studied two sets of inputs separately to understand their influence on the model: initial conditions (i.e., initial populations) and the model parameters.

Table 1 lists the results of the analysis of the initial conditions with different bounds: (50, 70) for plants, (10, 30) for males moths, (20, 40) for females moths and larvae, and (0, 5) for parasitoids. The total-order indexes are substantially larger than the first-order indexes, then higher-order interactions likely occur. In any case, is remarkable the influence of the initial population size of the male moths, and secondary the initial population size of the parasitoids.

Table 2 shows the influence of all set of parameters in our model. Focusing on the total-order indexes, the most important parameter was r_2 , the growth rate of the male moths. The interactions between plants with male moths, b_{12} , and plants with females moths and larvae, b_{13} , had a moderate influence, followed by the second and third order of parameters of the male moths, a_2 and c_2 .

4. Discussion

We used a system of differential equations based on a generalization of the model of García-Algarra et al. (2014) to simulate a complex biological system with facultative mutualism, obligate mutualism, and predation. These ecological interactions can be introduced in the model simply by modifying the sign of the parameters. Since biological parameters are easily identifiable in this model, we have effectively simulated an ecological system of two species that shift along a mutualism-antagonism interaction continuum, and then predicted the role of a third species (parasitoid) on the stability of the system.

Previous works have suggested that parasitoids may act as a regulatory mechanism that prevents populations of pollinating-seed predators from growing arbitrarily and thus, harming plant populations to the extinction threshold (Force and Thompson, 1984; Crabb and Pellmyr, 2006; Nunes et al., 2018; Elzinga and Bernasconi, 2009). However, this is based on observational studies and it had never been explored with experimental populations or by modeling approach. The effectiveness of the parasitoid as regulatory mechanism depends not only on its population size, but on the intensity of harming it causes to the nursery pollinators. In our model, these interaction was mediated by parameter b_{34} , the number of moths killed by each parasitoid, per

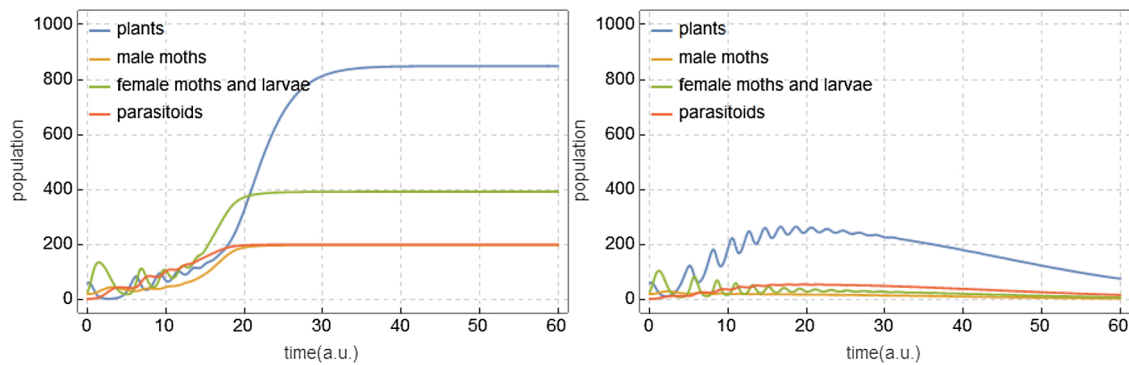


Fig. 6. Population dynamics of the system with two different predatory effects of the parasitoids on moth larvae. On the left, $b_{34} = -0.03$, and stationary stable populations are $X_1 = 850$, $X_2 = 198$, $X_3 = 392$ and $X_4 = 200$. on the right, $b_{34} = -1.83$ and stationary stable populations are all 0. Initial populations were set to $X_1(0) = 60$, $X_2(0) = 20$, $X_3(0) = 30$, $X_4(0) = 2$.

Table 1

Global sensitivity analysis of the initial conditions. In **1st** column we show the first-order sensitivity index, in **1st CI** the confidence interval for the first-order, in **Total** and **Total CI** the total-order indexes and their confidence level, respectively.

Populations	1st	1st CI	Total	Total CI
Plant	-0.05	0.07	0.04	0.06
Males	0.79	0.21	0.87	0.20
Females	0.00	0.00	0.00	0.00
Parasitoids	-0.02	0.14	0.18	0.11

Table 2

Global sensitivity analysis of the parameters in our model. In **1st** column we show the first-order sensitivity index, in **1st CI** the confidence interval for the first-order, in **Total** and **Total CI** the total-order indexes and their confidence level, respectively.

Parameters	1st	1st CI	Total	Total CI
r_1	-0.01	0.01	0.01	0.02
r_2	0.58	0.24	0.83	0.21
r_3	0.00	0.00	0.00	0.00
r_4	0.00	0.00	0.00	0.00
a_1	-0.00	0.00	0.00	0.00
a_2	0.05	0.07	0.10	0.12
a_3	-0.00	0.00	0.00	0.00
a_4	0.00	0.00	0.00	0.00
c_1	0.01	0.01	0.00	0.00
c_2	0.07	0.10	0.14	0.11
c_3	-0.02	0.04	0.02	0.01
c_4	0.00	0.00	0.00	0.00
b_{12}	0.09	0.10	0.21	0.16
b_{13}	-0.04	0.06	0.20	0.11
b_{21}	-0.01	0.01	0.01	0.02
b_{23}	0.03	0.06	0.02	0.03
b_{31}	0.00	0.00	0.00	0.00
b_{32}	-0.00	0.00	0.00	0.00
b_{34}	0.00	0.00	0.00	0.00
b_{43}	-0.00	0.00	0.00	0.00

unit of time. This parameter was carefully estimated from existing data, considering the developmental time of moth larvae, the probability of coincidence of larvae and parasitoids through year, the number of larvae visited by each parasitoid, and a factor to consider the coexistence of more than one species of parasitoids in the system. The changes in b_{34} modulated the duration of the oscillatory regimes and the stationary populations of the three species, as seen in Fig. 6. When this parameter increases, oscillations may provoke extinction of the female moths, causing the collapse of the entire system. In our example, we have modeled only one parasitoid species and assumed that its effect on the moths is strong, but it is common to observe more than one

parasitoid species coexisting (Elzinga et al., 2007). It is interesting to note that parasitoids may control over-exploitation of plants by moths simply by increasing larvae mortality, but the regulating processes can be subtler. In the Caryophyllaceae-Hadena-Microplitis system, experimental studies have reported that moth larvae rarely die after the injection of the wasps eggs, but the mean size reached by parasitized larvae is lower compared to healthy larvae (Elzinga et al., 2003). If the growth rate of parasitized caterpillars is constricted, the potential damage to the plants reproductive success is also limited. This effect could be assumed with parameter b_{13} , thus the flexibility of this population dynamics model permits parametrization of these and any other regulatory mechanisms to test their limits.

The analysis of global sensitivity of our model showed two remarkable points: (1) the initial population size of male moths, but also of parasitoids, had a higher effect on the stability of the system than the initial population size of plants and female moths, and (2) the stability also depends on the parameters related mainly with male moths (growth rate, interaction strength with plants, intraspecific competition and relative presence). This fact has already been highlighted by Labouche and Bernasconi (2010), who noted that in the *Silene latifolia-Hadena bicruris* system the female moths mostly carry with the antagonist side of the interaction (oviposition of predatory larvae). By contrast, male adult insects only contribute to the mutualist side because they provide pollination benefits without predatory costs, so they can support the net benefits of the interaction with the plant and therefore the population stability of the whole system. However, the pollination behavior of adult males in many other nursery pollination systems needs to be confirmed, and quantified if they occur, to explore properly the benefits and costs of the interaction Labouche and Bernasconi (2010). In our model, the separation of sexes of the pollinating-seed predator allowed us to analyze the male-derived benefits for the first time, compared with previous studies that modeled the stability of mutualisms by interspecific population regulation (Holland and DeAngelis, 2001, 2006; Westerbergh and Westerbergh, 2001).

As stated before, some nursery pollinated plants perform random or selective abortion of fruits, which has been modeled as an effective evolutionary strategy to regulate populations of their mutualist-antagonist partners (Holland and DeAngelis, 2001, 2006; Westerbergh and Westerbergh, 2001). However, the fruit abortion in response to *Hadena* moths has been described only in one species of Caryophyllaceae, *Silene latifolia* (Jolivet and Bernasconi, 2006). Our results demonstrate that the evolution of this mechanism may not be necessary if there exist other ecological processes of regulation mediated by third organisms. Pairs of species involved in nursery pollination systems do not interact in isolation, but are entangled into ecosystems. The presence of one or more species of parasitoids and other legitimate pollinators, in combination with the role of each sex on the net outcome of the interaction, and the spatio-temporal variation in species abundance,

will determine the direction of these mutualistic-antagonistic gradients, and confer more robustness to the system.

Conflict of interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

Appendix A. The role of male plants in the *Silene*-*Hadena* system

We can demonstrate that there is no influence of dividing the *Silene* population into males and females by taking equations (2) to (5), set the original plant population as female and add a male plant population, namely X_5 . Focusing only on X_1 and X_5 , since others populations and their dynamics remain the same, we have:

$$\frac{\dot{X}_1}{X_1} = (r_1 + b_{12}X_2 + b_{13}X_3 + b_{15}X_5) - (a_1 + c_1(b_{12}X_2 + b_{13}X_3 + b_{15}X_5))X_1, \quad (\text{A.1})$$

$$\frac{\dot{X}_5}{X_5} = (r_5 + b_{51}X_1) - (a_5 + c_5(b_{51}X_1))X_5. \quad (\text{A.2})$$

The only change is thus the new male plant population X_5 , which obviously behaves as an obligate mutualist of the female population, X_1 . There are only two stationary points for the male plant population, $X_5 = 0$ and $X_5 = (r_5 + b_{51}X_1)/(a_5 + c_5b_{51}X_1)$. For the first one, we return to the original system of equations (2) to (5). For the second one, we obtain a function of X_1 that shifts from a negative value, if $|r_5| > b_{51}X_1$ (we established that $r_5 < 0$), to a positive one. Thus, the unique dynamics that X_5 offers is going from extinction to its carrying capacity, depending on X_1 . On the other hand, the dynamics of X_1 depends on itself and b_{15} , since the last parameter modulates the influence of X_5 . The value of b_{15} cannot be too high compared to the other b_{1j} , because that would ensure the survival of X_1 and X_5 without pollinators. As a matter of fact, one may even argue that $b_{15} \sim 0$ because the reproduction of the plants depends on the pollination, a process that depends mainly on the moths and other flower visitors. Thus, if $b_{15} > 0$ but small enough, the influence of X_5 would be also very small and the system would behave as if X_5 does not exist.

Appendix B. Numerical values used on the simulations

We present the numerical values that we used in the simulations presented in the figures (Table B.3).

We calculated the value of each parameter by considering the following data and assumptions. First, all data were calculated using 1 day as the time unit and 1 year as the reproductive cycle. Since some individuals had more than one reproductive cycle per year, we considered this in our calculations. For some parameters that could not be obtained from actual data, reasonable estimates were used (Banks et al., 2017; Moxnes and Hausken, 2007).

Table B.3

Numerical values used in the simulations shown in Figs. 3 and 4. For Fig. 5, we change $r_1 + 0.1$. For a_i and c_i we did not use empirical values. For a_i we consider a carrying capacity of 4000, 200, 800 and 250 individuals for the plants, male moths, female moths (including larvae) and parasitoids, respectively. We considered $c_1 = a_1$ and $c_4 = a_4$, assuming that the relative presence of plants and parasitoids correspond to their carrying capacities. For c_2 and c_3 , we redistributed this value to equaling them, assuming that the saturation with respect to trophic interactions is similar.

Parameters	Numerical values	Units
r_1	−0.00274	time ^{−1}
r_2	−0.1	time ^{−1}
r_3	−0.0667	time ^{−1}
r_4	−0.0667	time ^{−1}
b_{12}	0.0527	(moths.time) ^{−1}
b_{13}	−0.0259	(moths.time) ^{−1}
b_{23}	0.0055	(moths.time) ^{−1}
b_{31}	0.0395	(plants.time) ^{−1}
b_{32}	0.0055	(moths.time) ^{−1}
b_{34}	−0.122	(parasitoids.time) ^{−1}
b_{43}	0.011	(moths.time) ^{−1}
a_1	0.00025	(plants.time) ^{−1}
a_2	0.005	(moths.time) ^{−1}
a_3	0.00125	(moths.time) ^{−1}
a_4	0.004	(parasitoids.time) ^{−1}
c_1	0.00025	1
c_2	0.0025	1
c_3	0.0025	1
c_4	0.004	1

Table B.4

Variables used to calculate the parameters of our model, as indicated in B.5. * They counted 13.72 flowers visited by 18 males and 17.25 flowers visited by 19 females over 40 minutes. ** *Silene* individuals produce an average of 6 flowers per week. + *Hadena* larvae consume approximately 5 fruits along the month it takes to turn into adults. + + We consider the probability of coincidence of 2 generations of male moths, which last 10 days each over 6 months with another 2 generations of female moths, which last 15 days over the same six months. This gives a probability of $2 * 10/180 * 2 * 15/180 = 0.0185$. * We need to consider an additional factor because we only model one parasitoid species and there are at least 13 species; the rounding is ours. ** We assume 2 generations of parasitoids, which last 15 days each over 6 months with another 2 generations of moth larvae, which last 30 days over the same 6 months. This gives $2 * 15/180 * 2 * 30/180 = 0.0556$. Most data come from original studies centered on the system *Silene latifolia-Hadena bicruris-Microplitis tristis* (¹ L. Giménez-Benavides, personal observation; ² (Brantjes, 1976); ³ (Elzinga et al., 2002); ⁴ J.L. Yela, personal communication; ⁵ (Elzinga et al., 2007); ⁶ (Labouche and Bernasconi, 2010); ⁷ (Wright and Meagher, 2004); ⁸ (Blair and Wolfe, 2004); ⁹ (Burkhardt et al., 2009); ¹⁰ (Elzinga and Bernasconi, 2009))

Variable	Description	Units	Estimate
lt_1	Mean lifetime of <i>Silene</i> individuals	day	365 (¹)
lt_2	Mean lifetime of male <i>Hadena</i> individuals	day	10 (^{2,4})
lt_3	Mean lifetime of female <i>Hadena</i> individuals	day	15 (^{2,3,4})
lt_4	Mean lifetime of parasitoids	day	15 (⁵)
v_{fm}	Flowers visited by male moths*	flowers/min.moth	0.019 (⁶)
v_{ff}	Flowers visited by female moths*	flowers/min.moth	0.043 (⁶)
t_{ac}	Active time of moths on a day	min/days	120 (^{1,4,6})
rt_{fpp}	Ratio of flowers produced daily per plant**	flowers	0.857 (⁷)
rt_{ffv}	Ratio of fertilized flowers per visit		0.8 (⁶)
rt_{fov}	Ratio of fertilized ovules in a fruit		0.45 (⁶)
n_{ov}	Ovules per flower		300 (⁶)
rt_{ssu}	Ratio of seed survival		0.1
rt_{sge}	Ratio of seed germination		0.5 (⁸)
rt_{egg}	Ratio of visited flowers with egg oviposition		0.9 (⁶)
n_{me}	Moth eggs laid per visit		10 (⁶)
n_{sf}	Seeds per fruit		135 (⁶)
n_{fl}	Fruits ate by larvae per day+	fruits/day	0.167 (³)
n_{ef}	Eggs available for each female moth		100 (^{2,3,4})
pb_{mf}	Ratio of male and female moths coincidence + +		0.0185 (^{2,3,4})
rt_{egf}	Ratio of egg fertilization		0.35
rt_{meh}	Ratio of moth egg hatching and larval establishment		0.5 (¹⁰)
rt_{lsu}	Ratio of larvae survival to adult		0.75 (¹⁰)
rt_{fm}	Ratio of moth males vs females		0.5 (⁴)
lt_{ml}	Mean time of moth larvae development	day	30 (³)
n_{pe}	Parasitoid eggs laid per visit		18 (⁵)
f_{pav}	Factor of parasitoid predatory influence*		20 (⁵)
v_{lp}	Larvae visited by parasitoids	1/days.parasitoid	1 (⁵)
pb_{pl}	Ratio of larvae and parasitoids coincidence**		0.0556 (^{3,5})
rt_{peh}	Ratio of parasitoid egg hatching		0.8 (⁵)
rt_{psu}	Ratio of parasitoid survival to adult		0.25

Table B.5

Calculations used to estimate parameters (B.3) with the data available (B.4). We used the denominator g_2 because moths and parasitoid species may have up to two generations per year (Elzinga 2007). For b_{31} we used 150/365 because *Silene* individuals produce flowers for approximately five effective months, from May to October. For b_{23} and b_{32} we used the mean value of lt_2 and lt_3 to avoid unobserved asymmetries among male and female moths.

Parameter	Calculation
g_2	$= (0.5 * 365 + 0.5 * 180)$
r_i	$= lt_i^{-1}$
b_{12}	$= v_{fm} t_{ac} (rt_{fpp}^{-1}) rt_{ffv} rt_{fov} n_{ov} rt_{ssu} rt_{sge} / g_2$
b_{13}^+	$= v_{ff} t_{ac} (rt_{fpp}^{-1}) rt_{ffv} rt_{fov} n_{ov} rt_{ssu} rt_{sge} / g_2$
b_{13}^-	$= b_{13}^+ rt_{egg} n_{me} n_{sf} n_{fl} rt_{meh} rt_{lsu}$
b_{13}	$= b_{13}^+ - b_{13}^-$
b_{23}	$= n_{ef} (\frac{lt_2 + lt_3}{2}) pb_{mf} rt_{egf} rt_{meh} rt_{lsu} rt_{fm} / g_2$
b_{31}	$= rt_{fpp} rt_{ffv} (n_{fl}^{-1}) rt_{lsu} (150/365)$
b_{32}	$= n_{ef} (\frac{lt_2 + lt_3}{2}) pb_{mf} rt_{egf} rt_{meh} rt_{lsu} (1 - rt_{fm}) / g_2$
b_{34}	$= lt_{ml} f_{pav} v_{lp} pb_{pl} / g_2$
b_{43}	$= n_{pe} lt_4 v_{lp} pb_{pl} rt_{peh} rt_{psu} / g_2$

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